AFRILANG Stdlib

std/c/gamemcts

Français. Bibliothèque AFRILANG ‘std/c/gamemcts‘ (niveau complex, catégorie complex). Elle expose 7 fonction(s) : uctSelect, rollout, simulate, bestAction, updateStats, treePolicy, mctsValue.
‘import ”std/c/gamemcts”‘ · ‘use gamemcts‘
- ‘uctSelect(wins list number, visits list number, totalVisits number, explore number) → number‘ - ‘rollout(state list number, depth number) → number‘ - ‘simulate(state list number, sims number) → number‘ - ‘bestAction(wins list number, visits list number) → number‘ - ‘updateStats(wins list number, visits list number, action number, reward number) → list number‘ - ‘treeP **English.** AFRILANG library ‘std/c/gamemcts‘ (tier complex, category complex). Exports 7 function(s): uctSelect, rollout, simulate, bestAction, updateStats, treePolicy, mctsValue.
‘import ”std/c/gamemcts”‘ · ‘use gamemcts‘
- ‘uctSelect(wins list number, visits list number, totalVisits number, explore number) → number‘ - ‘rollout(state list number, depth number) → number‘ - ‘simulate(state list number, sims number) → number‘ - ‘bestAction(wins list number, visits list number) → number‘ - ‘updateStats(wins list number, visits list number, action number, reward number) → list number‘ - ‘treePolicy(state l

Catégorie / Category	Modules complex (c/)
Niveau / Tier	complex
Fonctions / Functions	7

June 16, 2026

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/gamemcts' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : uctSelect, rollout, simulate, bestAction, updateStats, treePolicy, mctsValue.

'import "std/c/gamemcts"' · 'use gamemcts'

```
- `uctSelect(wins list number, visits list number, totalVisits number, explore number
    ) → number`
- `rollout(state list number, depth number) → number`
- `simulate(state list number, sims number) → number`
- `bestAction(wins list number, visits list number) → number`
- `updateStats(wins list number, visits list number, action number, reward number) →
    list number`
- `treePolicy(state list number, actions list number) → number`
- `mctsValue(wins list number, visits list number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/gamemcts"
use gamemcts
```

```
Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés — graphes, NLP, réseaux complexes.
```

3. Référence API

```
•
 uctSelect(wins list number, visits list number, totalVisits number, explore number)
    → number•
 rollout(state list number, depth number) → number•
 simulate(state list number, sims number) → number•
 bestAction(wins list number, visits list number) → number•
 updateStats(wins list number, visits list number, action number, reward number) →
    list•
 treePolicy(state list number, actions list number) → number•
 mctsValue(wins list number, visits list number) → number
```

4. Exemples d'utilisation

```
import "std/c/gamemcts"
use gamemcts
```

```
create result = uctSelect(/* args */)
say result
```

```
create result = rollout(/* args */)
say result
```

```
create result = simulate(/* args */)
say result
```

```
create result = bestAction(/* args */)
say result
```

```
create result = updateStats(/* args */)
say result
```

```
create result = treePolicy(/* args */)
say result
```

5. Bonnes pratiques

```
•
 Préférez `import "std/c/gamemcts"` en tête de fichier.•
 Lancez `afrilang check` avant de livrer.•
 Combinez avec les modules stdlib de la même catégorie.•
 Voir /stdlib/ pour le source live et les démos playground.•
 Signalez les problèmes sur GitHub si une export manque.
```

6. Annexe — source

module gamemcts

```
function uctSelect(wins list number, visits list number, totalVisits number, explore
    number) returns number
end
```

```
function rollout(state list number, depth number) returns number
end
```

```
function simulate(state list number, sims number) returns number
end
```

```
function bestAction(wins list number, visits list number) returns number
end
```

```
function updateStats(wins list number, visits list number, action number, reward
    number) returns list number
end
```

```
function treePolicy(state list number, actions list number) returns number
end
```

```
function mctsValue(wins list number, visits list number) returns number
end
end
```

English

1. Overview

AFRILANG library ‘std/c/gamemcts‘ (tier complex, category complex). Exports 7 function(s): uctSelect, rollout, simulate, bestAction, updateStats, treePolicy, mctsValue.
‘import ”std/c/gamemcts”‘ · ‘use gamemcts‘

```
- `uctSelect(wins list number, visits list number, totalVisits number, explore number
    ) → number`
- `rollout(state list number, depth number) → number`
- `simulate(state list number, sims number) → number`
- `bestAction(wins list number, visits list number) → number`
- `updateStats(wins list number, visits list number, action number, reward number) →
    list number`
- `treePolicy(state list number, actions list number) → number`
- `mctsValue(wins list number, visits list number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/gamemcts"
use gamemcts
```

```
Tier: complex · Category: Complex modules (c/)
Advanced domains — graphs, NLP, complex networks.
```

3. API reference

```
•
 uctSelect(wins list number, visits list number, totalVisits number, explore number)
    → number•
 rollout(state list number, depth number) → number•
 simulate(state list number, sims number) → number•
 bestAction(wins list number, visits list number) → number•
 updateStats(wins list number, visits list number, action number, reward number) →
    list•
 treePolicy(state list number, actions list number) → number•
 mctsValue(wins list number, visits list number) → number
```

4. Usage examples

```
import "std/c/gamemcts"
use gamemcts
```

```
create result = uctSelect(/* args */)
say result
```

```
create result = rollout(/* args */)
say result
```

```
create result = simulate(/* args */)
say result
```

```
create result = bestAction(/* args */)
say result
```

```
create result = updateStats(/* args */)
say result
```

```
create result = treePolicy(/* args */)
say result
```

5. Best practices

```
•
 Prefer `import "std/c/gamemcts"` at the top of your file.•
 Run `afrilang check` before shipping.•
 Combine with related stdlib modules from the same category.•
 See /stdlib/ for the live source and playground demos.•
 Report issues on GitHub if an export is missing or undocumented.
```

6. Appendix — source

module gamemcts

```
function uctSelect(wins list number, visits list number, totalVisits number, explore
    number) returns number
end
```

```
function rollout(state list number, depth number) returns number
end
```

```
function simulate(state list number, sims number) returns number
end
```

```
function bestAction(wins list number, visits list number) returns number
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function updateStats(wins list number, visits list number, action number, reward
    number) returns list number
end
```

```
function treePolicy(state list number, actions list number) returns number
end
```

```
function mctsValue(wins list number, visits list number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/gamemcts"`
- Vérification : `afrilang check`
- Documentation : `https://afrilang.dev/stdlib/std/c/gamemcts/`

Source (extrait)

```
module gamemcts

function uctSelect(wins list number, visits list number, totalVisits number, explore
    number) returns number
end

function rollout(state list number, depth number) returns number
end

function simulate(state list number, sims number) returns number
end

function bestAction(wins list number, visits list number) returns number
end

function updateStats(wins list number, visits list number, action number, reward
    number) returns list number
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function treePolicy(state list number, actions list number) returns number
end

function mctsValue(wins list number, visits list number) returns number
end
end
```